INVESTMENTS

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CHAPTER 3

How Securities Are Traded

Study Guide · Prepared for Personal Use

LO1 · How Firms Issue Securities

Investment bankers manage a firm's sale of securities in the **primary market.** Once issued, investors trade those securities among themselves in the **secondary market** — a transaction (e.g., selling your Bombardier shares to another investor) that has no effect on the total number of shares outstanding.

Privately Held Firms

A privately held company is owned by a small number of shareholders and has fewer disclosure obligations, saving money and protecting competitively sensitive information. In Canada, many private firms are family-controlled subsidiaries of larger multinationals and represent a significant part of the economy. Because their shares do not trade on public exchanges, they are **illiquid,** and investors demand a price concession to compensate. Firms wishing to raise funds privately can use a **private placement** — selling directly to a small number of institutional or wealthy investors without the cost of a full public registration. A newer option, **equity crowdfunding** (permitted in Canada since 2015 under provincial prospectus exemptions), lets external investors acquire small stakes through a dedicated web portal.

Going Public — Underwriting

A firm's first sale of shares to the public is its **initial public offering (IPO);** a later sale of additional shares by an already-public firm is a **seasoned new issue.** Public offerings are marketed through **underwriting** by investment dealers (in Canada, often subsidiaries of chartered banks, e.g., Nesbitt Burns/BMO, Scotia McLeod/Scotiabank). A lead firm forms an **underwriting syndicate** to share the risk. In a **firm commitment (bought deal),** underwriters buy the securities from the issuer at the offering price less a spread, then resell to the public — bearing full risk if shares don't sell. In a **best-efforts agreement,** the banker helps sell the issue without assuming that risk.

The registration statement, once approved by the provincial securities commission, becomes the **prospectus.** Canada's **Short Form Prospectus Distribution System (SFPD)** (the equivalent of the U.S.'s shelf registration) lets firms with existing annual disclosure file only minor supplementary updates for a new issue — approved in days rather than weeks.

IPO Mechanics — Bookbuilding and Underpricing

Before pricing, underwriters run **road shows** to gauge investor interest; large investors' indications of interest form a **book,** and the process is called **bookbuilding.** Investors have an incentive to reveal true interest because share allocations reward those who show strong demand — in turn, underwriters must price the offering attractively for early participants, which routinely leads to **underpricing:** a price jump on the first day of public trading.

> **Real examples:** Twitter's 2013 IPO closed its first day up 73% from the $26 offer price. Alibaba's 2014 IPO closed up 38%. Canada's Jamieson Wellness (2017) closed its first day up 9.84% from its $15.75 offer price. While explicit IPO costs run about 7% of funds raised, underpricing is a much larger implicit cost — money "left on the table" that the firm could otherwise have raised.

LO2 · How Securities Are Traded

Financial markets evolve to meet traders' needs. Four market structures represent increasing levels of organisation:

Market Type	Key Features
Direct Search Markets	Least organised. Buyers and sellers must find each other directly (e.g., a used-refrigerator classified ad). Sporadic participation, low-value/non-standard goods.
Brokered Markets	Brokers offer search services once trading activity justifies specialisation (e.g., real estate). The primary market for new securities is itself a brokered market — investment bankers act as brokers seeking investors.
Dealer Markets	Dealers trade for their own account, profiting from the bid-ask spread. Save traders search costs since posted prices are easy to find. Most bonds trade over-the-counter in dealer markets.
Auction Markets	Most integrated: all traders converge (physically or electronically) to transact, avoiding the need to search across dealers. The NYSE is a classic example; participants can arrive at mutually agreeable prices and save the bid-ask spread.

Types of Orders

Market orders execute immediately at the current price. Example: if Enbridge's best bid is $49.32 and best ask is $49.40, a market buy fills near $49.40 and a market sell near $49.32 — an $0.08 bid-ask spread. Large orders may fill at multiple prices if they exceed the posted quote's depth.

Price-contingent orders specify the price at which the investor is willing to trade. A **limit buy order** executes only at or below a specified price; a **limit sell order** only at or above. A **stop order** (stop-loss or stop-buy) is triggered once the price crosses a threshold, converting into a market order at that point — useful for limiting losses on both long and short positions.

LO3 · Securities Markets

Canada once had five stock exchanges; today, equity trading is consolidated on the **TSX** (Toronto Stock Exchange), with the **TSX Venture Exchange (TSXV)** serving smaller/emerging companies and the **Canadian Securities Exchange (CSE)** as an alternative listing venue. Derivatives trade on the Canadian Derivatives Exchange. In 2000, the TSX converted from a member-owned mutual organisation to a for-profit, publicly traded company — a trend called **demutualization** that has occurred at exchanges worldwide.

Trading has shifted overwhelmingly to **electronic** execution: in the U.S., electronic trading's share of equity volume rose from about 16% in 2000 to over 80% by the end of that decade, with even greater dominance elsewhere. Electronic Communication Networks (ECNs) let participants post buy/sell orders that are automatically crossed without a human intermediary.

The NYSE and NASDAQ

The **NYSE** historically operated with designated market makers ("specialists") at physical trading posts, but is now overwhelmingly electronic. **NASDAQ** has always been a pure dealer market: all trades pass through a dealer, and the bid-ask spread is embedded in the execution price rather than charged as a visible commission.

LO4 · Trading Costs

Trading costs have two components: an **explicit** commission and an **implicit** cost from the bid-ask spread (and any price concession for trading beyond the dealer’s posted quantity).

Full-Service vs. Discount Brokers

Full-service brokers execute orders, hold securities, extend margin loans, facilitate short sales, and provide research/advice — sometimes managing a **discretionary account** where the broker trades on the client’s behalf (raising the risk of **churning:** excessive trading to generate commissions). **Discount brokers** offer only the basic execution/custody services at a fraction of the cost — as low as $10 per trade versus $100–$300 at full-service firms. Facing this competition, many full-service brokers have shifted to fee-based **wrap accounts** charging 2.5–3.5% of assets annually instead of per-trade commissions.

The Bid-Ask Spread and Execution Quality

As commissions have fallen, the bid-ask spread has become a proportionally larger share of total trading cost. On markets with floor brokers, **price improvement** can occur when two public orders cross at a price between the quoted bid and ask, avoiding the full spread. In a pure dealer market like NASDAQ, every trade passes through the dealer and is subject to the spread — but that cost is embedded in the execution price and never itemised for the investor.

LO5 · Trading with Margin and Short Sales

Buying on Margin

Investors who buy on margin borrow part of the purchase price from their broker, who in turn borrows from a bank at the **call money rate** and charges the client that rate plus a service fee. Securities bought on margin are held in **street name** as collateral for the loan. In Canada, the **initial margin requirement** is 30% for the most marginable stocks (i.e., up to 70% may be borrowed), rising toward 100% for low-priced/riskier stocks; the U.S. initial requirement is 50%. If the percentage margin falls below the **maintenance margin** (30% in Canada), the broker issues a **margin call** requiring the investor to add cash or securities — or the broker may sell out the position. Margin calls can arrive with little notice: on October 25, 2000, when the TSX fell 8.3% in a day, some brokerages gave clients only hours to meet calls before selling out their accounts.

Short Sales

A short sale lets an investor profit from an expected price decline: the investor borrows shares (through the broker) and sells them, later **covering** the position by buying back shares to return to the lender. The proceeds of the sale are credited to the investor's account, and the broker requires additional collateral (margin) — typically 50% of the position's value at initiation in Canada, with a 30% maintenance margin. Because losses on a short position are theoretically unlimited (the price can rise indefinitely), short-sellers often pair the position with a **stop-buy order** to cap potential losses.

LO6 · Regulation of Securities Markets

Canadian securities regulation is primarily **provincial,** not federal — there is no single national securities regulator. The Ontario Securities Act (1945), Canada's first, has served as the template most other provinces followed. The federal government influences securities activity indirectly through the Canada Business Corporations Act and provisions of the Criminal Code.

Self-Regulatory Organisations (SROs)

- **IIROC —** the Investment Industry Regulatory Organization of Canada, the national SRO overseeing investment dealers and trading activity on debt and equity marketplaces (formed 2008 from the merger of the IDA and Market Regulation Services). Governs trading on the TSX, TSXV, and CSE.
- **CSA —** the Canadian Securities Administrators, an informal umbrella body of the provincial/territorial securities regulators that works to harmonise regulation across jurisdictions.
- **MFDA —** the Mutual Fund Dealers Association of Canada, the national SRO for the mutual fund industry.
- **CIPF —** the Canadian Investor Protection Fund, established by IIROC and the exchanges to protect investors (up to $1 million) if their brokerage firm fails — the securities-industry analogue to CDIC deposit insurance.

The Regulatory Rationale

Provincial securities commissions protect investors through **registration** of industry participants and by requiring **disclosure** of material facts, via the prospectus at issuance and ongoing quarterly/annual reports thereafter. Approval of a prospectus is *not* an endorsement of the investment's quality — regulators only ensure the relevant facts are disclosed; investors form their own judgment of value.

Provincial fragmentation creates real friction: a transaction between, say, Ontario investors and a Quebec-listed company can trigger overlapping jurisdiction and potentially duplicate investigations — though some argue this competitive multiplicity also benefits investors. By contrast, in the **United States,** the **SEC** (created by the Securities Exchange Act of 1934) provides unified national oversight of exchanges, OTC trading, brokers, and dealers, sharing responsibility with the CFTC (futures) and the Federal Reserve (overall financial system health).

Chapter 3 — Big-Picture Takeaways

- Firms raise capital in the primary market via underwriters; investors then trade in the secondary market, where trading structures range from informal direct search to fully electronic auction markets.
- IPOs are commonly underpriced — a real (if implicit) cost to the issuing firm, on top of explicit underwriting fees of roughly 7%.

- Trading costs combine an explicit commission and an implicit bid-ask spread; the spread's relative importance has grown as commissions have fallen.
- Margin buying and short selling both provide leverage and both carry margin-call risk if the position moves against the investor.
- Canadian securities regulation is provincially based and disclosure-focused; SROs (IIROC, MFDA) and the CIPF provide industry self-regulation and investor protection alongside provincial securities commissions.

Margin and short-sale mechanics are the most heavily tested quantitative content in this chapter.

F1 | Initial Percentage Margin (Buying on Margin)

Margin Ratio = (Market Value of Assets − Loan) / Market Value of Assets

= Equity in Account / Market Value of Assets

Equity is the investor's own contribution; the loan is what was borrowed from the broker. In Canada, the initial requirement for the most marginable stocks is 30% (up to 70% may be borrowed).

F2 | Margin Ratio After a Price Change

Margin Ratio = Equity in Account / Current Value of Stock

Equity moves dollar-for-dollar with the stock's value (the loan amount is fixed). As the stock price falls, equity shrinks faster in percentage terms than the price itself.

F3 | Price Triggering a Margin Call (Long Position)

Solve: (Shares × P − Loan) / (Shares × P) = Maintenance Margin for P

Set the margin ratio equal to the maintenance margin (30% in Canada) and solve for the critical stock price P. Below this price, a margin call is triggered.

F4 | Margin Ratio on a Short Sale

Margin Ratio = Market Value of Assets in Account / Value of Stock Owed

Assets = short-sale proceeds + posted collateral. "Value of stock owed" is the current market value of the shares that must eventually be bought back. Ratio starts at 150% or more if the initial margin requirement is 50%.

F5 | Price Triggering a Margin Call (Short Position)

Solve: Market Value of Assets / (Shares × P) = Maintenance Margin for P

Unlike a long position, a short-seller's margin ratio falls as the price RISES (since the value of stock owed grows while account assets stay fixed). Solve for the critical price above which a margin call is triggered.

F6 | Front-End Load Break-Even Return

Break-even return = Load / (1 − Load)

For a 6% front-end load, $1,000 buys only $940 of fund shares. Break-even = 60/940 = 6.4% — the cumulative return needed on the net investment just to recover the load. (Formalised further in Chapter 4.)

EXAMPLES 3.1 & 3.2 | Buying on Margin & the Margin Call Price

Setup: Investor buys 100 shares at $100/share ($10,000 total), paying $6,000 and borrowing $4,000 from the broker.

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Initial margin = ($10,000 − $4,000) / $10,000 = 60%
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If the price falls to $70/share:

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Stock value = $7,000;  Loan = $4,000 (unchanged);  Equity = $3,000
Margin ratio = $3,000 / $7,000 = 42.9%
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Price that triggers a margin call (30% maintenance minimum):

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(100P − 4,000) / 100P = 0.30   →   100P − 4,000 = 30P   →   P = $57.14
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If the price falls below $57.14, the investor receives a margin call.

EXAMPLE 3.4 | Short Sale Mechanics

Setup: Short-sell 1,000 shares of ACB at $100/share ($100,000 proceeds). Broker requires 50% margin; investor posts $50,000 in T-bills as collateral.

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Account assets = $100,000 (proceeds) + $50,000 (T-bills) = $150,000
Margin ratio = $150,000 / $100,000 = 150%
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If ACB falls to $70 (favourable):

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Cover cost = 1,000 × $70 = $70,000;  Profit = $100,000 − $70,000 = $30,000
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Price that triggers a margin call (130% maintenance minimum, i.e. 30% margin):

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$150,000 / 1,000P = 1.30   →   P = $115.38
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If ACB rises above $115.38, the investor receives a margin call. A stop-buy order (e.g., at $120) is commonly used to cap losses on a short position.

PRACTICE PROBLEM | Prairie Capital — A Trader's Day

Scenario

You are a trader at Prairie Capital in Calgary. Two clients bring you separate orders on the same day, both involving RockFord Energy (RFE), currently quoted at a bid of $54.80 and an ask of $55.10 on the TSX.

Part A — Orders and the Bid-Ask Spread

(i) What is the bid-ask spread on RFE, in dollars?

(ii) Client 1 places a market order to buy 200 shares. At approximately what price does this execute?

(iii) Client 2 places a limit sell order at $56.00. Will it execute immediately? Explain.

Part B — Buying on Margin

Client 3 wants to buy 500 shares of RFE at the $55.10 ask price, using the Canadian initial margin requirement of 30%.

(i) How much of the purchase price must Client 3 pay in cash, and how much can be borrowed?

(ii) Calculate the initial percentage margin.

(iii) If RFE falls to $40.00, calculate the new equity and margin ratio.

(iv) At what price will Client 3 receive a margin call, given a 30% maintenance margin?

Part C — Short Selling

Client 4 is bearish on RFE and shorts 1,000 shares at $55.10, posting the 50% required margin in T-bills.

(i) What are the short-sale proceeds, and how much collateral (T-bills) must Client 4 post?

(ii) Calculate the initial margin ratio on the short position.

(iii) If RFE rises to $63.00, will Client 4 receive a margin call under a 30% maintenance margin? Show your work.

(iv) What order type would you recommend Client 4 use to limit losses, and why?

Part D — Market Structure & Regulation

(i) Is the TSX best classified as a dealer market or an auction market? Explain.

(ii) If Client 3's broker fails and cannot return their margin securities, what organisation protects the client, and up to what limit?

(iii) Prairie Capital wants to launch a new stock offering for a private client's firm. Briefly describe the primary-market process from registration through to the prospectus.

Hints & Guidance

Part A tests LO2. Part B tests LO5 and Formulas F1–F3 (mirror Example 3.1/3.2 exactly, just with new numbers). Part C tests LO5 and Formulas F4–F5 (mirror Example 3.4). Part D synthesises LO3, LO6, and LO1.